WEEK 8 · LECTURE L1 · UNIT III

Dashboard Design

What an executive should see in ten seconds, and what to leave out.

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AFTER THIS LECTURE YOU SHOULD BE ABLE TO

What this lecture prepares you for

- Apply the ten-second test: hierarchy, restraint and the right chart.
- Distinguish strategic, operational and analytical dashboards.
- Explain dashboard theater and the counter-metric discipline.
- Apply Few, Tufte, Cleveland and McGill, and working memory research to dashboard design.

COURSE OUTCOMES

CLO 4 (metrics, KPIs and dashboards).

THE TEN-SECOND TEST

Can an executive grasp the message in ten seconds?

Hierarchy
The most important number is largest and first.

Restraint
Few tiles, little decoration, nothing without a decision behind it.

The right chart
Lines for change over time; bars for comparison.

DASHBOARD TYPES

Match the dashboard to its user

Type	For whom	What it shows
Strategic	Executives	A few KPIs, monthly or quarterly
Operational	Managers	Daily work, updated often
Analytical	Analysts	Data to explore and filter

EXAM LENS

Do not mix types on one screen. A CHRO screen full of operational detail fails all three users.

DASHBOARD THEATER

Pair every target with a counter-metric

Target	Counter-metric that reveals gaming
Chatbot deflection	Resolution rate
Time to fill	Quality of hire
Course completion	Assessment scores and on-the-job behavior
Low attrition	Regretted exits among high performers

EXAM LENS

A number that becomes a target changes behavior (Goodhart, Week 3). The counter-metric exposes gaming.

EXERCISE

Critique a five-tile CHRO dashboard

Overall attrition

Lagging. Counter: regretted exits among high performers.

Share of staff under weakly rated managers

Leading. Counter: exit rate within that group.

Active self-service use

Leading. Counter: helpdesk tickets per employee.

Frontline digital skills completion

Leading. Counter: assessment scores.

Hiring adverse impact ratio

Compliance. Counter: time to fill.

Your task

Which tile would you remove, and which decision does each remaining tile serve?

THEORY IN DEPTH · FEW (2006)

A dashboard is a single screen for monitoring

THE CORE IDEA

A visual display of the most important information needed to achieve objectives, on one screen, seen at a glance.

HOW IT WORKS

Four rules follow: most important only, one screen, glanceable, tied to objectives.

LIMITS AND CRITIQUE

Some argue interactive drill-downs make the single-screen rule outdated.

IN PRACTICE

If a CHRO needs to scroll, the dashboard has already failed Few's test.

EXAM LENS

Quote Few's definition precisely; it is the standard reference.

THEORY IN DEPTH · TUFTE (1983)

Maximize the data-ink ratio

THE CORE IDEA

Most ink on a chart should show data; decoration that carries no data is chartjunk.

HOW IT WORKS

Erase non-data ink and redundant ink; keep only what informs.

LIMITS AND CRITIQUE

Some studies show embellishment can aid memory, so minimalism is not always best.

IN PRACTICE

3D pie charts and gradient tiles on an HR dashboard add ink and remove clarity.

EXAM LENS

Present the critique of strict minimalism to show balance.

THEORY IN DEPTH · CLEVELAND AND MCGILL (1984)

Some encodings are read more accurately than others

THE CORE IDEA

People judge position along a common scale best, then length, then angle, then area, then color saturation.

HOW IT WORKS

Choose bar and dot charts for comparisons; avoid pies and bubbles where precision matters.

LIMITS AND CRITIQUE

Based on laboratory tasks; real dashboards also need speed and memorability.

IN PRACTICE

Show attrition by location as sorted bars, not as a pie.

EXAM LENS

Use the ranking to justify a chart choice explicitly.

THEORY IN DEPTH · MILLER (1956); COWAN (2001)

Working memory holds very little

THE CORE IDEA

Miller suggested about seven items; Cowan's later review put the limit nearer four chunks.

HOW IT WORKS

Beyond a few items, viewers stop comparing and start skimming.

LIMITS AND CRITIQUE

Chunking and familiarity raise effective capacity.

IN PRACTICE

Five tiles is already near the limit; fourteen tiles guarantee that nothing is compared.

EXAM LENS

Cite Cowan as the update to Miller; it shows awareness of how evidence evolved.

PRACTICE FOR THE END-TERM EXAMINATION

Questions to rehearse

Using Few, Tufte and Cleveland and McGill, critically evaluate the design of HR dashboards and propose principles for a CHRO dashboard. **20 marks**

A strong answer: applies each theory; covers hierarchy, restraint and encoding; adds counter-metrics; notes limits of minimalism.

"Dashboard theater" — explain the concept and how counter-metrics address it. **20 marks**

A strong answer: links to Goodhart and Campbell; gives target and counter-metric pairs; evaluates residual risks.

Why should a CHRO dashboard contain few tiles? **10 marks**

A strong answer: uses working memory research (Miller; Cowan) and Few's definition.

CARRY THESE INTO THE EXAMINATION

Three things to remember

1. **Pass the ten-second test: hierarchy, restraint, right chart**
2. **Pair every target with a counter-metric**
3. **Fewer tiles, each tied to a decision**

SET READINGS · FROM THE LESSON PLAN

Read for this lecture

- Kavanagh, M. J., & Johnson, R. D. (2018), Ch 14 (HR metrics, workforce analytics, reporting and HR dashboards).

COMPANION

Each set reading is explained in the matching Unit Reader on rksingh.tech/materials.

Sources for this lecture

Works named on the slides of this lecture. They are background sources, not set readings.

Cleveland, W. S., & McGill, R. (1984). Graphical perception: Theory, experimentation, and application to the development of graphical methods. *Journal of the American Statistical Association, 79*(387), 531–554. https://doi.org/10.1080/01621459.1984.10478080

Cowan, N. (2001). The magical number 4 in short-term memory: A reconsideration of mental storage capacity. *Behavioral and Brain Sciences, 24*(1), 87–114. https://doi.org/10.1017/s0140525×01003922

Few, S. (2006). *Information dashboard design: The effective visual communication of data*. O'Reilly.

Kavanagh, M. J., & Johnson, R. D. (Eds.). (2018). *Human resource information systems: Basics, applications, and future directions* (4th ed.). Sage.

Miller, G. A. (1956). The magical number seven, plus or minus two: Some limits on our capacity for processing information. *Psychological Review, 63*(2), 81–97. https://doi.org/10.1037/h0043158

Tufte, E. R. (1983). *The visual display of quantitative information*. Graphics Press.